

MEMORY

Sensory memory and short-term memory

Course instructor:

Dr Shweta Chawak

Assistant Professor

Humanities and Social Sciences

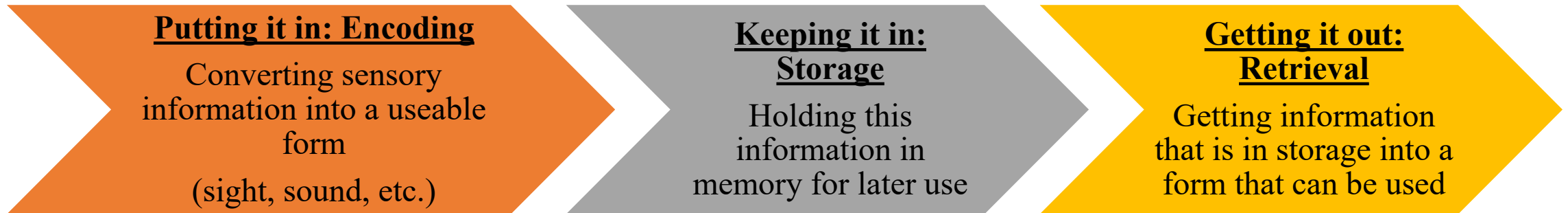
IIT Bombay



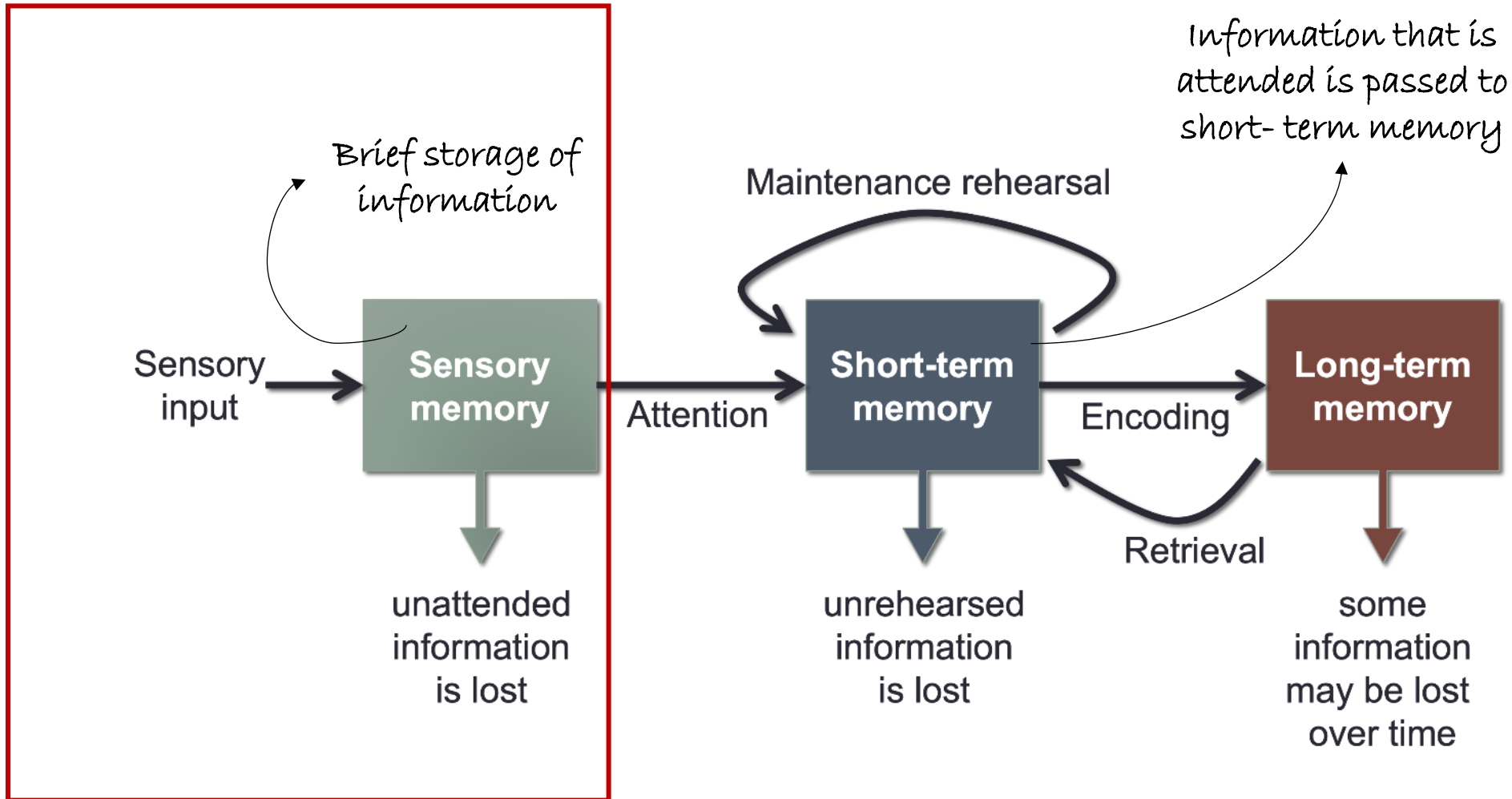
What is Memory?

Memory is an active system that receives information from the senses, puts that information into a usable form, organizes it as it stores it away, and then retrieves the information from storage

Three processes of memory

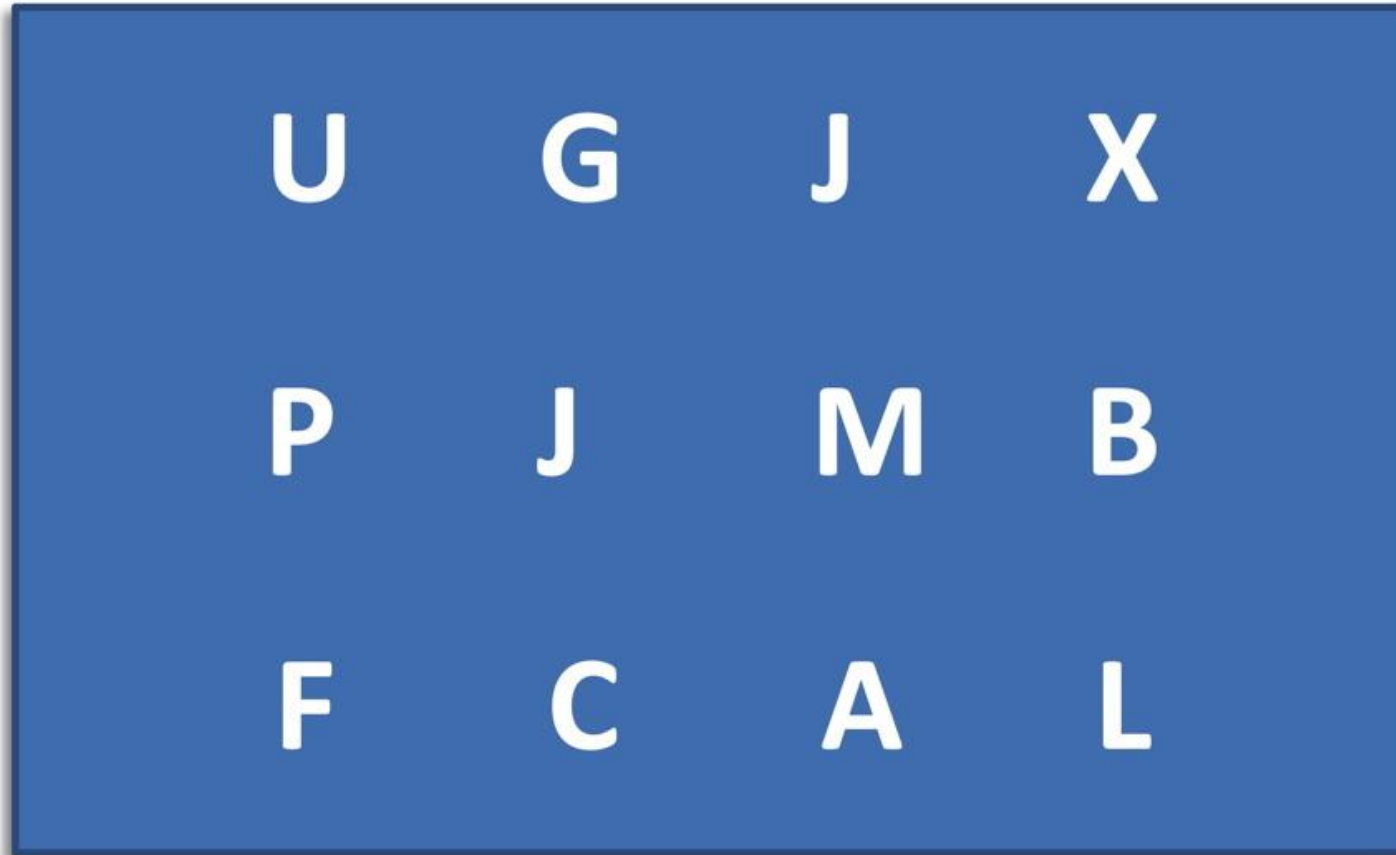


Information processing model

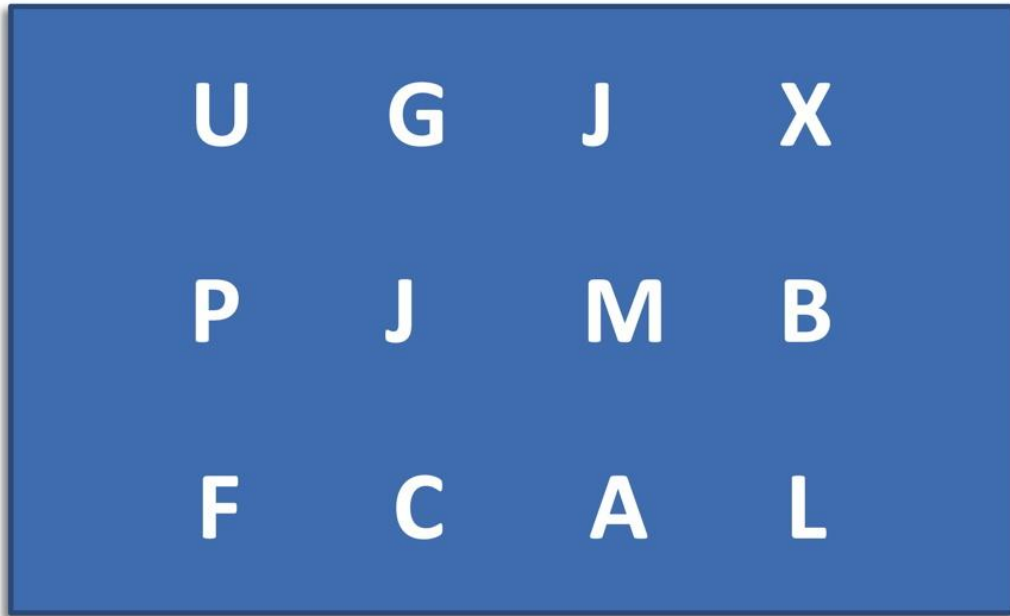


Pay attention to the next slide...

How do we store memories?! Sensory Memory

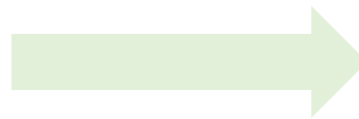


How do we store memories?! Sensory Memory



Sensory memory:

- raw information from the senses is held
- the very first stage of memory
- very brief period of time (within a second)



Why Sensory memory

- Follow a conversation
- Read smoothly
- Perceive motion as continuous
- Understand spoken language
- Recognize visual scenes
- Experience the world as stable and coherent

Iconic memory

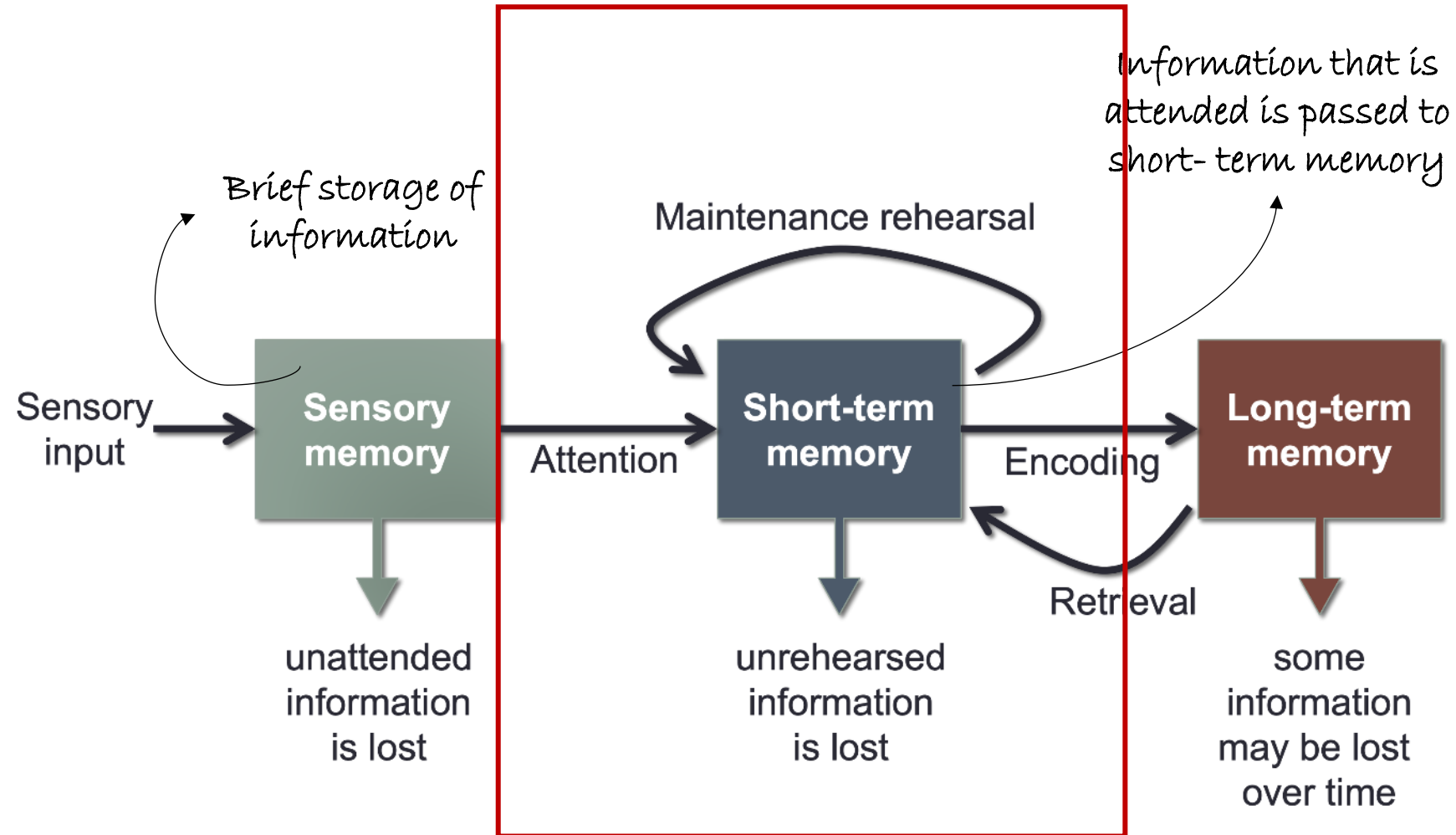
- visual sensory memory
- (fraction of a second)

Echoic memory

- Auditory sensory memory
- (2-4 seconds)



Information processing model



The Digit-span test

<https://www.psychtoolkit.org/experiment-library/digitspan.html>



How do we store memories?! Short-term Memory

Short-term memory (STM):

- Information is held for brief periods of time while being used
 - Also called working memory
 - Mostly encoded in auditory form
 - Lasts around 30 seconds
-
- George Miller (1956)
 - Digit-span test: memory test
 - Magical number 7, plus or minus two: 7 ± 2
 - Selective Attention: Focusing (voluntarily) on a selected portion of sensory input (e.g., selective hearing; cocktail-party effect)

(Atkinson & Shiffrin, 1968)



How do we store memories?! Short-term Memory

Maintenance rehearsal

- Practice of saying some information to be remembered over and over in one's head in order to maintain it in short-term memory
- Interference causes memory loss
- STM hindered when it exceeds capacity
 - Chunking helps remember more information

E.g.,

- Reading a definition or fact again and again right before a quiz, but not linking it to other information or personal experiences
- Memorizing a locker combination by repeating it constantly until opening the lock
- Chunking: 99-22-53-41...



RECAP

- Memory is an active system that receives information from the senses, puts that information into a usable form, organizes it as it stores it away, and then retrieves the information from storage
 - Putting it (Encoding)
 - Keeping in storage (holding information)
 - Getting it out (Retrieval)
- Information processing model
 - Sensory memory – short term memory – long-term memory
 - Sensory memory: raw sensory information hold [iconic (visual) & echoic (audio)]
 - Short term memory: information held for shorter duration (7 ± 2)
- How to hold information: maintenance rehearsal or chunking